

Graham Freifeld

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Education

- PhD, computer science University of Edinburgh, 2022 - 2026
Advisor: Professor Heng Guo
Thesis title: New sampling algorithms for acceleration, derandomization, and non-crossing spanning trees
- BS, mathematics and computer science UC Santa Cruz, 2018 - 2022

Research Interests

- Algorithms and complexity for counting and sampling problems
- Counting and reconfiguration of planar structures (triangulations, non-crossing spanning trees, non-crossing matchings)

Articles

- Simulating Gaussian boson sampling on graphs in polynomial time
Konrad Anand, Zongchen Chen, Mary Cryan, Graham Freifeld, Leslie Ann Goldberg, Heng Guo, and Xinyuan Zhang
arXiv: 2511.16558
- Sink free orientations: a local sampler with applications
Konrad Anand, Graham Freifeld, Heng Guo, Chunyang Wang, and Jiaheng Wang
doi: 10.4230/LIPIcs.APPROX/RANDOM.2025.60
arXiv: 2502.05877
- Rapid mixing of the flip chain over non-crossing spanning trees
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, Mark Jerrum, and Jiaheng Wang
doi: 10.4230/LIPIcs.SoCG.2025.8
arXiv: 2409.07892
- Approximate counting for spin systems in sub-quadratic time
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, and Jiaheng Wang
doi: 10.46298/theoretics.25.3
arXiv: 2306.14867

Seminars/Workshops

- Jerrumfest, a workshop on algorithms and complexity
Queen Mary University, July 2025

- Scottish Combinatorics Meeting,
Postgraduate Combinatorial Conference
University of Glasgow, April 2025
- Workshop on the Combinatorial, Algorithmic, and Probabilistic aspects of Partition
Functions
University of Amsterdam, March 2025
- ADYN Summerschool on Algorithms, Dynamics, and Information Flow in Networks
Dortmund University, July 2024
- Cambridge Algorithms and Complexity Workshop
Cambridge University, April 2024

Teaching

- Marker (Grader), University of Edinburgh
Winter 2025 - INFR08026 Introduction to Algorithms and Data Structures
Winter 2024 - INFR11020 Algorithmic Game Theory
Fall 2023 - INFR11201 Randomized Algorithms
- Tutor, University of Edinburgh
Fall 2023 - INFR11201 Randomized Algorithms
- Tutor, University of California at Santa Cruz
Spring 2022 - CSE 102 Introduction to Analysis of Algorithms
Winter 2022 - Math 111A Algebra
Fall 2021 - CSE 102 Introduction to Analysis of Algorithms
Winter 2021 - CSE 102 Introduction to Analysis of Algorithms

Work experience

- Assistant research technician, earth sciences division
Lawrence Berkeley National Laboratory
Berkeley, California, USA
Summer 2018, 2021